An orthogonal Matching Pursuit Algorithm Optimization Design Based On FPGA Implementation

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Abstract: According to compression sensing reconstruction algorithm of Orthogonal Matching Pursuit (OMP) algorithm the problem of each iteration can't select the optimal atomic,to optimize the OMP algorithm design, ensures that each iteration of the current allowance minimum observation signal,and proposes a kind of optimize the OMP algorithm based on FPGA to realize the hardware structure design.In the matrix decomposition part adopts modified Cholesky decomposition methods, avoid root operation, to reduce the calculation time delay,easy to FPGA implementation.The whole system adopts parallel computing,resource reuse technology,improve the computing speed and reduce resource utilization.In the Quartus II development environment for the design of the RTL description,on the FPGA simulation platform for simulation, the simulation results verify the validity of the design.

Key words: compressed sensing；orthogonal matching pursuit；modified Cholseky decomposition；FPGA

1 INTRODUCTION

Traditional signal acquisition and processing mainly use the Nyquist Sampling Theorem realization, while the Nyquist Sampling Theorem requires sampling frequency shall be not less than twice the highest frequency of the spectrum of analog signal which makes the high frequency signal acquisition limited greaty.[1] Along with the rapid development of information technology,Signal bandwidth has been increase dramatically, with the industry enter the age of big data, the requirement of signal processing ability and hardware device became harder gradually,which brought greatly challenges to data processing[2].

In recent years, there was proposed a compressive sensing theory，it has the advantages of dealing data acquisition and compression processing in the same time, using less data recover the original signal accurately, so as to make the sampling frequency get rid of constraint by Nyquist sampling and working in the lower sampling frequency.Compressed sensing theory including three aspects as follow:signal sparse representation, the structure of the measurement matrix, signal reconstruction algorithm design.Signal reconstruction algorithm design is key to compression sensing theory research.This article is based on greed iterative[3]matching pursuit algorithm.In orthogonal matching pursuit[4] represented in terms of greedy algorithm and its improved algorithm has the advantages of signal reconstruction speed,small computational complexity compared with other algorithms [5].

Compressed sensing signal reconfiguration hardware design is still in the preliminary stage, there are still exist many issues to researching and scholars have been publish a series of achievement. References[6] designed compressed sensing signal reconstruction algorithm in method of VLSI(Very Large Scale Integration).In references[7],the OMP algorithm hardware design perform resources multiplexing under specific order, which improved resource utilization and running speed . References[8] introduce the LU decomposition method based on FPGA,its can obtain good acceleration ability on computer platform.However, there are a large number of matrix multiplication and division in the processing of LU decomposition, it occupied the FPGA hardware resources great quantities and lead calculation delayed. In this paper, using the modified Cholesky decomposition method in the period of matrix decomposition, it avoid the root square operation, decrease the times of multiplication and make the operation speed faster.

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2 ORTHOGON ALMATCHING PURSUIT ALGORITHM

In the processing of compressed sensing sampling, the original signal, $x \in R^N$,sparse degree is K (K<< N).Designing a $M \times N$ (M < < N) random measurement matrix,the column vector of random measurement matrix (l = 1, 2,..., N) called atoms.According to the theory of compressed sensing, the sparse signal projected on the random measurement matrix to get a shorter length $M \times 1$ observation vector $y \in R^M$ than original signal.Matching pursuit algorithm core is select best matching atoms with current observation signal margin r (initialized to observation signal y) in the Nth times of iteration from random measurement matrix Φ .Sent the selected atoms add to the candidate subset Γ to form new candidate subset. According to the candidate subset, we calculate new estimate signals $\hat{x}$ and new observation signal margin r . In the next iteration, we continue to select the best match with the observed signal margin of atoms to form a new candidate subset that calculate $\hat{x}$ and r till the iteration end.

3 MATCHING PURSUIT ALGORITHM OPTIMIZATION DESIGN

3.1 Orthogonal Matching Pursuit Algorithm Atom Selection Criteria

Assumption that choose a subset f_{i_l} from the over-complete atom set$\{ f_i$,*l=1,2,...,n* $\}$,defined as $V_1 = f_{i_1}$,$V_{n+1} = V_n \oplus f_{i_{n+1}}$,which $\oplus$ represents the space's orthogonal sum. W_{n+1} is the orthogonal complement space of V_{n+1} and V_n .Defined $P_{V_{n+1}}$ is the orthogonal projection operator of V_{n+1} , $P_{V_{n+1}} = P_{V_n} + P_{W_{n+1}}$,where $f_{i_{n+1}} \in V_{n+1}$ .Meanwhile we defined ϕ_{n+1} as the orthogonal projection operator of $f_{i_{n+1}}$ on the W_{n+1} . ϕ_{n+1} can be equivalent to the following formula.

$$\begin{aligned}\Phi_{n+1} &= P_{W_{n+1}} f_{i_{n+1}} \\ &= (P_{V_{n+1}} - P_{V_n}) f_{i_{n+1}} \\ &= P_{V_{n+1}} f_{i_{n+1}} - P_{V_n} f_{i_{n+1}} \\ &= f_{i_{n+1}} - P_{V_n} f_{i_{n+1}} \end{aligned} \tag{1}$$

Normalizing ϕ_{n+1} in The formula (1), $\tilde{\phi}_{n+1} = \phi_{n+1} / \|\phi_{n+1}\|$,marking $\phi_1 = \tilde{\phi}_1 = f_{i_1}$ .In addition, making the function a_{i_n} satisfied the following formula （K=1,2,…,n）.

$$a_{i_1}^{(1)} = f_{i_1}$$

$$a_{i_k}^{n+1} = a_{i_k}^n - \tilde{\Phi}_{n+1} / \|\Phi_{n+1}\| \left\langle f_{i_{n+1}}, a_{i_k}^n \right\rangle$$

$$a_{i_{k+1}}^{n+1} = \tilde{\Phi}_{n+1} / \|\Phi_{n+1}\| \tag{2}$$

Utilizing the conclusion of reference [8],we can get the measured value y of orthogonal projection formula on the V_{n+1} as follows.

$$P_{V_{n+1}} y = \sum_{k=1}^{n+1} \left\langle a_{i_k}^{n+1}, y \right\rangle f_{i_n} \tag{3}$$

Obtaining orthogonal projection coefficient of y on the V_{n+1}（*K=1,2,...,n*）on the basis of formula（3）.

$$D_{i_1}^1 = \left\langle f_{i_1}, y \right\rangle$$

$$D_{i_k}^{n+1} = D_{i_k}^n - \left\langle a_{i_k}^n, f_{i_{n+1}} \right\rangle \left\langle \tilde{\Phi}_{n+1} / \|\Phi_{n+1}\|, y \right\rangle$$

$$D_{i_{n+1}}^{n+1} = \left\langle \tilde{\Phi}_{n+1} / \|\Phi_{n+1}\|, y \right\rangle \tag{4}$$

Untill n+1 iteration, $\|r^{n+1}\|^2 = \|y\|^2 - \left\langle P_{V_{n+1}} y, y \right\rangle$.

Under the condition of the y 's measured value fixed, in order to minimize observation signal margin r which equivalent to the $\left\langle P_{V_{n+1}} y, y \right\rangle$ maximized..By the formulas (1) to (4) will be as follows.

$$\begin{aligned}\left\langle P_{V_{n+1}} y, y \right\rangle &= \left\langle P_{V_n} y, y \right\rangle + \frac{\left| \left\langle f_{i_{n+1}}, y \right\rangle - \left\langle P_{V_n} f_{i_{n+1}}, y \right\rangle \right|^2}{\|\Phi_{n+1}\|^2} \\ &= \left\langle P_{V_n} y, y \right\rangle + \frac{\left| \left\langle f_{i_{n+1}}, y \right\rangle - P_{V_n} y \right|^2}{1 - \left\langle f_{i_{n+1}}, P_{V_n} f_{i_{n+1}} \right\rangle} \\ &= \left\langle P_{V_n} y, y \right\rangle + \frac{\left| \left\langle f_{i_{n+1}}, r^n \right\rangle \right|^2}{1 - \left\langle f_{i_{n+1}}, P_{V_n} f_{i_{n+1}} \right\rangle} \end{aligned} \tag{5}$$

$$C_{n+1} = \frac{\left| \left\langle f_{i_{n+1}}, r^n \right\rangle \right|^2}{1 - \left\langle f_{i_{n+1}}, P_{V_n} f_{i_{n+1}} \right\rangle} \tag{6}$$

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where the measured value $\left\langle P_{V_n} y, y \right\rangle$ fixed,which equivalent to the C_{n+1} maximized. OMP atom selected criteria $\lambda^n = \arg\max_{l=1,2,\cdots,N} \left| r^{n-1}, f_l \right|$,it became as following after optimized.

$$\lambda^n = \arg\max_{l=1,2,\cdots,N} C_l \tag{7}$$

3.2 Orthogonal Matching Pursuit Algorithm Optimization Calculation Steps

Analyzing the Orthogonal Matching Pursuit algorithm atom selection criteria.The calculation step of optimization of the OMP algorithm reconstruction algorithm shown as follows.

Step1：Initializing $\hat{x}=0$， $r^0=0$， $\Gamma^0=\varnothing$， $n=1$

Step2: Selecting the best matching atom index λ^n of the current observation signal margin r^{n-1} .

$$\lambda^n = \arg\max_{l=1,2,\cdots,N} C_l \tag{8}$$

Step3:Updating the candidate subset $\Gamma^n = \Gamma^{n-1} \cup \lambda^n$. Meanwhile,Recording the reconstruction atoms collection in the sensing matrix.

$$\tilde{\Phi}_n = [\tilde{\Phi}_{n-1}, f_{\lambda^n}] \tag{9}$$

Step4:Using the existing atoms approach to the original signal.

$$\hat{x}_n = \arg\min \left\| y - \tilde{\Phi}_n \hat{x} \right\|_2 \tag{10}$$

Step 5: Updating the residual value.

$$r_n = y - \tilde{\Phi}_n \hat{x}_n \tag{11}$$

Step 6:$n=n+1$,if n < k, then returns to step 2,stop iteration until obtain the approximate signal $\hat{x}$.

3.3 Orthogonal Matching Pursuit Algorithm Hardware Structure Optimization Design

The optimization OMP algorithm can be divided into four modules.The first module matching reconstruction algorithm to calculation step 2.Through the selection of atoms, we can obtain the most matching residual atoms after using the formula (1) to (5) .

The second module matching reconstruction algorithm to calculation step 3.Updating the candidate subset Γ ,we can obtain the augmented matrix $\tilde{\Phi}_n$.

The third module matching reconstruction algorithm to calculation step 4.To solve the problems of l_0 minimum norm model and least squares method,we can obtain the estimated value of the original signal. Calculating the augmented matrix $\tilde{\Phi}$ of inverse method to obtain the original estimated value of $\hat{x}$. However, matrix $\tilde{\Phi}$ is a non-square matrix , calculating non-square matrix is generally used Moore-pseudo method.

The Moore-pseudo of matrix $\tilde{\Phi}$ can be expressed as.

$$\tilde{\Phi}^+ = (\tilde{\Phi}^T \times \tilde{\Phi})^{-1} \times \tilde{\Phi}^T \tag{12}$$

In the formula (15),making $G = \tilde{\Phi}^T \times \tilde{\Phi}$ .The above problems are directly transformed into calculate inverse of G . $G \in R^{t \times t}$ is a symmetric positive definite matrix,which can not calculate on FPGA easily. so we could start the matrix decomposition of matrix G , then calculate its inverse matrix.

There are several matrix factorizations, including QR decomposition[9],singular value decomposition,LU decomposition,Cholesky decomposition.The calculation processes of QR decomposition and singular value decomposition are complex, it's not easy implement on FPGA.There are a large number of matrix multiplications and divisions in the processing of LU decomposition,which occupied the FPGA hardware resources great quantities and lead calculation delayed .Although, in the Cholesky decomposition calculation exist the square root operation delay and division calculation,its less complexity than LU decomposition of matrix. For the delay problem of root operation in decomposition calculation,the Cholesky decomposition can be used to avoid the delay problem of rooting operation. This calculation formula of the modified Cholesky decomposition shown as follows.

$$G = L \times D \times L^T \tag{13}$$

$$L_{i,j} = \frac{1}{D_{i,j}} \left(G_{i,j} - \sum_{t=1}^{i-1} L_{i,j} \times L_{j,t} \times D_{t,t} \right) \tag{14}$$

$$D_{i,j} = \left(G_{i,j} - \sum_{t=1}^{i-1} L_{i,j} \times L_{j,t} \times D_{t,t} \right) \tag{15}$$

According to formulas (13) to (15),we obtain the inverse matrix of G .

$$G^{-1} = (L^{-1})^T \times D^{-1} \times L^{-1} \tag{16}$$

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According to formula (16),we obtain the calculation formula of the original signal estimated value $\hat{x}$.

$$\hat{x} = (\tilde{\Phi}^T \times \tilde{\Phi})^{-1} \times \tilde{\Phi}^T \times y$$
$$= G^{-1} \times \tilde{\Phi}^T \times y \quad (17)$$

The fourth module matching reconstruction algorithm to calculation step 5.Calculating residual r,it's preparing for the next iteration.

4 OPTIMIZE OMP ALGORITHM BASED ON FPGA IMPLENTATION

The hardware circuit is mainly composed of four modules, which divided into two parts.

The concrete circuit diagram shown as figure 1.

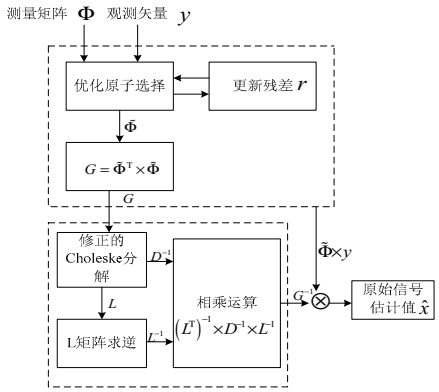


Figure 1 Hardware circuit diagram of optimize OMP algorithm

The first part including the given two inputs--measurement matrix Φ and observation vector y respectively.Initializing the residual r by inputting observation vector $y \in R^N$.The length of each column matrix is N,Designing N multipliers and N-1 adders which working in parallel and finished in one work clock cycle.Meanwhile, Calculation of measurement matrix Φ and the residual r are completed within one clock respectively.Using multiple register stored observation vector y,and measurement matrix value Φ stored in several RAM.We can calculate the atomic index λ^n with using optimization atom selection criterion obtained from formula (1) to (6).Updating the subset of candidate in step 3, we can obtain reconstructed matrix $\tilde{\Phi}$.Solving this formula $G = \tilde{\Phi}^T \times \tilde{\Phi}$ and obtaining matrix G.

The second part is a process of inversion of matrix G.It consists of a hardware circuit of Cholesky decomposition, inversion of matrix L and multiplication hardware circuit.Through the triangle matrix L and diagonal matrix D, we can obtain modified Cholesky decomposition matrix G,To avoid square operation,the formula is shown as follows.

$$G = L \times D \times L^T \quad (18)$$

The relationship between matrix L and D is correlation dependence, calculate its element under specific order . Finally,we can get G^{-1} by following formula:

$$G^{-1} = (L^{-1})^T \times D^{-1} \times L^{-1} \quad (19)$$

5 SIMULATION AND VERIFICATION

This work compared the restructure experiment results between optimized OMP algorithm and standard OMP algorithm by one-dimension.Making an assumption that the length of sparse signal x is N=256,sparse coefficient k=6,optimized OMP algorithm and standard OMP algorithm adopt Gauss Random measurement matrix $\Phi \in R^{M \times N}$,then,we record restructure success rate of previous two algorithms,it shown as following figure 2.

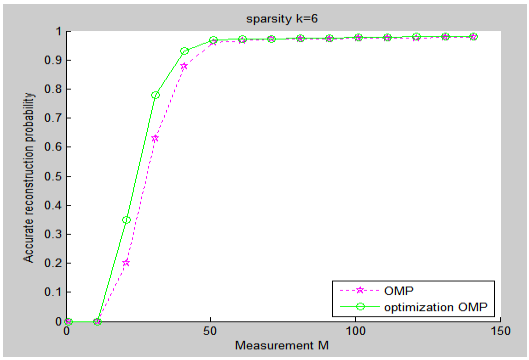


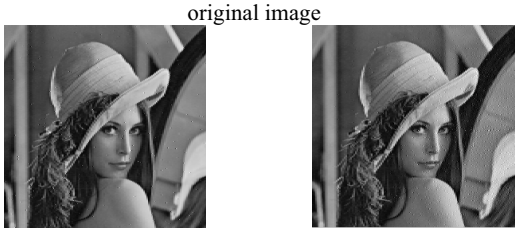
Figure 2 The relationship between the success rate and the measurement frequency under the Gaussian matrix

This work test the effectiveness of optimized OMP algorithm by two-dimension image signal experiment. We selected the classic experiment image Lena of 256 pixels by 256 pixels as the experiment scale and, adopting Gaussian random matrix as a measurement matrix. This paper make a reconstruction experiment contrastive research into the standard OMP algorithm and optimize OMP algorithm which the Sampling frequency M/N=50%. Reconstruction results shown as following figure 3.

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OMP algorithm optimize OMP algorithm

figure 3 Reconstruction results of two algorithm

When the sampling rate M/N=50%,signal-noise ratio of optimization OMP and standard OMP algorithms is 34.53dB and 33.72dB respectively.As a result that reconstruction precision of the optimized OMP algorithm is higher than the standard OMP algorithm.

Using the Modelsim software to simulate hardware circuit of the optimized OMP algorithm.The simulation result shown in Figure 4 and Figure 5.

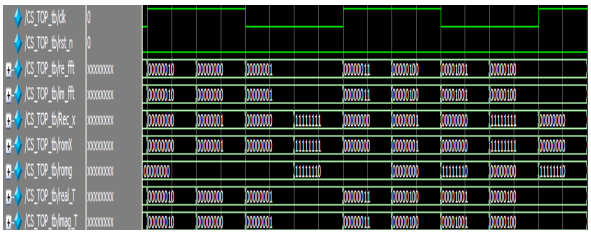


Figure 4 Modelsim functional simulation

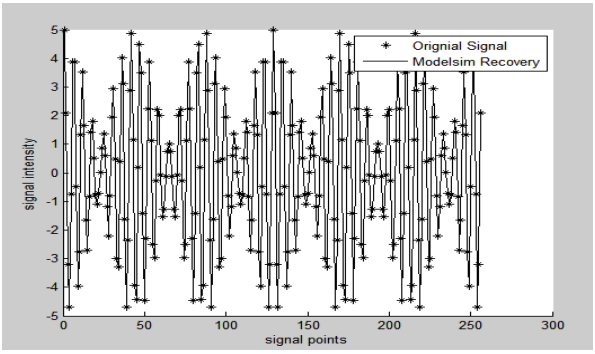


Figure 5 The figure of Modelsim simulation results contrast

6 CONCLUSION

In this paper, OMP algorithm choose optimal atoms in a short time by optimizing atoms choice principle,it shorten reconfiguration times and improve speed of algorithm reconstruction.Meanwhile,we design the hardware structure of FPGA,which keep a balance between occupied resources and running time. This design adopts hardware description language Verilog HDL to make an achievement of optimization OMP reconstruction algorithm and integrates algorithm by utilizing Quartus software.According to making a Co-Simulation to Register Transfer Level description by Matlab and Modelsim,we get a better reconstruction result which indicates optimized OMP algorithm can recover original signal in a short time.

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